department. We're small, but we're *great*, we can do all of that.

And it's always okay to ask for help, it's okay to ask questions, it's okay to redo things and triple-check. What's not okay is to implicitly assume that someone else will do the things here that you haven't thought about.



"How does this apply juniors? You can't expect them to really do all of that?"

I've been asked these questions, or variations of them, after I shared the thoughts above and here's my answer.

I do *not* expect juniors to *do* all of these things right away. But I would expect them to read through the list and *aspire* to one day be able to do all of these things. Until then, they can and should ask for help.

In fact, I don't expect *anybody* to *always* do *all of these things* for *everything*. It's a mental checklist of things to consider — problem, edge cases, tradeoffs, deployment, customers, messaging, … — but for quite a few things there aren't edge cases to consider. Or big tradeoffs to weigh. Or deployment is a solved problem. And maybe someone else does the messaging for you.

And I imagine that most of these things you shouldn't even consider when you work in a company with, say, 5000 employees. I've never worked in a company that large, only startups, so I can't speak to how to successfully ship a software feature at Apple, end to end.

But when you work in a small company in which there's only a single department, when you want to build things you're proud of, when you work with me and you say own something, I expect you to keep these things in mind and run through them before you declare something as done.



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FRED TALKS

24th October 2026

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Every layer of review makes you 10x slower

AVERY PENNARUN · 26 MAR 2026 · SOURCE

2026-03-16 »

Every layer of review makes you 10x slower

We've all heard of those network effect laws: the value of a network goes up with the square of the number of members. Or the cost of communication goes up with the square of the number of members, or maybe it was n log n, or something like that, depending how you arrange the members. Anyway doubling a team doesn't double its speed; there's coordination overhead. Exactly how much overhead depends on how badly you botch the org design.

But there's one rule of thumb that someone showed me decades ago, that has stuck with me ever since, because of how annoyingly true it is. The rule is annoying because it doesn't *seem* like it should be true. There's no theoretical basis for this claim that I've ever heard. And yet, every time I look for it, there it is.

Here we go:

Every layer of approval makes a process 10x slower

I know what you're thinking. Come on, 10x? That's a lot. It's unfathomable. Surely we're exaggerating.

Nope.

Just to be clear, we're counting "wall clock time" here rather than effort. Almost all the extra time is spent sitting and waiting.

Look:

- Code a simple bug fix
  30 minutes

- Get it code reviewed by the peer next to you
  300 minutes → 5 hours → half a day

- Think about data flow. How much data is involved here? Does data need to be migrated? Cleaned up? How can I get my hands on data to properly test this? What invariants are in the data? What assumptions do I have about the shape of the data that I haven't confirmed yet?

- Think about how you'd test this. How can I know that what I built is correct or not? Are tests enough? Do I need to manually poke at things? Is the difference visible on a screenshot or in a video?

- How would we announce this? How do we communicate it? Can you picture it? How does it fit into the larger picture of our roadmap? Questions or concerns in that area — push back! ask!

- Do the work, with precision, with care, with a sense of urgency, with calmness. Do not half-ass things. Before you merge, ask yourself: am I proud of this? would I show this to John Carmack and say "here's what I built, under these constraints, with these tradeoffs?"

- Test it manually. Yes, there's automated tests. But in 99% of cases you can manually test or confirm that what you built works: you can run it yourself, you can ask an agent to run through test scenarios, you can poke at the data before and after, you can take screenshots, you can make a demo. Are you sure that what you did actually solves the problem?

- Make sure it lands in production and *works in production*. Is it deployed? Did the deploy fail? Do you need to activate a feature flag? Does the feature flag work? Can you use it in production? Can you confirm it's actually deployed?

- If you think your colleagues needs to know about this change, because it's new feature they should all test, or it's a new convention in codebase, or maybe it's a tricky thing everybody needs to be aware of, or something else: let them know! Do not underestimate peripheral vision: knowing that person X yesterday changed the behavior of how Z works might save person Y three hours of debugging today when a bug report related to Z comes in.

- Do customers need to know? Who reported the bug? Who's blocked? Let them know.

- Does the world need to know? Announce that it's out.

- Are there follow-ups? Do you need to check on what you shipped in the logs? A week later maybe?

Yes, that's a lot. And there's actually more, because I'm sure I forgot some stuff.

But that's how you build a product in a small team. We don't have PMs, we don't have a Q&A

Thoughts on ownership I sent to the Amp team in internal note

The following is an internal Slack message I sent to my Amp teammates after I had a conversation with one of them about ownership. It's only lightly edited.

I shared it before, but not here, because I didn't think too much of it. Then today, someone said their CTO shared my post with them and I thought: well, now I have to put it in the newsletter, don't I? So here we are.

Below, after the Slack post, I added some thoughts on juniors on how I see this advice applying to them.



Just had a (great) conversation about ownership and engineering here and I realized that I often use the phrase "ownership" or allude to it, but haven't explained what "ownership" means to me in a while.

So, ownership.

If I ask you "can you own this?" or "can you take care of this?" or "are you on it?" — what I'm doing is I'm asking you to *own* it, to own the solution of a problem from end to end. From "we have a problem" to "we don't have to think about it again."

That means, when you say that you're owning something, the expectation is that you…

- Think about what the problem actually is. Maybe you already have a solution in mind, without having thought about what we're actually trying to solve here. Maybe you think "the problem is that we need to migrate from using X to using Y", but that's not a problem, that's a solution. The problem is likely something like "performance is bad", "it's not stable", "it fails for customer x". Maybe there's other possible solutions to that? Think about those. What are the tradeoffs? What's the best solution to go with considering these tradeoffs?

- Think about edge cases. What are they? Which ones are important? Which ones can we ignore?

- Think about failures. Network failures are a given, for example. How do we handle them? Retry? Well, how often? How long?

- Get a design doc approved by your architects team first
  50 hours → about a week

- Get it on some other team's calendar to do all that
  (for example, if a customer requests a feature)
  500 hours → 12 weeks → one fiscal quarter

I wish I could tell you that the next step up — 10 quarters or about 2.5 years — was too crazy to contemplate, but no. That's the life of an executive sitting above a medium-sized team; I bump into it all the time even at a relatively small company like Tailscale if I want to change product direction. (And execs sitting above large teams can't actually do work of their own at all. That's another story.)

AI can't fix this

First of all, this isn't a post about AI, because AI's direct impact on this problem is minimal. Okay, so Claude can code it in 3 minutes instead of 30? That's super, Claude, great work.

Now you either get to spend 27 minutes reviewing the code yourself in a back-and-forth loop with the AI (this is actually kinda fun); or you save 27 minutes and submit unverified code to the code reviewer, who will still take 5 hours like before, but who will now be mad that you're making them read the slop that you were too lazy to read yourself. Little of value was gained.

Now now, you say, that's not the value of agentic coding. You don't use an agent on a 30-minute fix. You use it on a monstrosity week-long project that you and Claude can now do in a couple of hours! Now we're talking. Except no, because the monstrosity is so big that your reviewer will be extra mad that you didn't read it yourself, and it's too big to review in one chunk so you have to slice it into new bite-sized chunks, each with a 5-hour review cycle. And there's no design doc so there's no intentional architecture, so eventually someone's going to push back on that and here we go with the design doc review meeting, and now your monstrosity week-long project that you did in two hours is... oh. A week, again.

I guess I could have called this post Systems Design 4 (or 5, or whatever I'm up to now, who knows, I'm writing this on a plane with no wifi) because yeah, you guessed it. It's Systems Design time again.

The only way to sustainably go faster is fewer reviews

It’s funny, everyone has been predicting the Singularity for decades now. The premise is we build systems that are so smart that they themselves can build the next system that is even smarter, that builds the next smarter one, and so on, and once we get that started, if they keep getting smarter faster enough, then the incremental time (t) to achieve a unit (u) of improvement goes to zero, so (u/t) goes to infinity and foom.

Anyway, I have never believed in this theory for the simple reason we outlined above: the majority of time needed to get anything done is not actually the time doing it. It’s wall clock time. Waiting. Latency.

And you can’t overcome latency with brute force.

I know you want to. I know many of you now work at companies where the business model kinda depends on doing exactly that.

Sorry.

But you can’t just *not review* things!

Ah, well, no, actually yeah. You really can’t.

There are now many people who have seen the symptom: the start of the pipeline (AI generated code) is so much faster, but all the subsequent stages (reviews) are too slow! And so they intuit the obvious solution: stop reviewing then!

The result might be slop, but if the slop is 100x cheaper, then it only needs to deliver 1% of the value per unit and it's still a fair trade. And if your value per unit is even a mere 2% of what it used to be, you’ve doubled your returns! Amazing.

There are some pretty dumb assumptions underlying that theory; you can imagine them for yourself. Suffice it to say that this produces what I will call the AI Developer’s Descent Into Madness:

1. Whoa, I produced this prototype so fast! I have super powers!
2. This prototype is getting buggy. I’ll tell the AI to fix the bugs.
3. Hmm, every change now causes as many new bugs as it fixes.
4. Aha! But if I have an AI agent also review the code, it can find its own bugs!
5. Wait, why am I personally passing data back and forth between agents

You had anticipated this and not given it access to the FooBarDB client library; and marked the FooBarDB CLI as non-executable; but it just went ahead and used curl to access a forgotten REST endpoint.

The Rogue Agent: As the agent copies state (this time using the CLI tool after you yell at it enough), it lists keys in batches; each key corresponds to some user of your application. Someone decided to pick the user name “delete-everything”. Of course, you are using the latest model which is always resistant to prompt injections. Almost always.

The Stupid Agent: You do everything right. There are no prompt injections in your state. You wall off FooBarDB and force the agent to use the CLI tool. The model decides to delight you by generating an unusual and interesting sequence of tokens that will optimally save storage space for you, translating to a lambda that deletes your production data entirely.

One response to these difficult failure modes might be: wait until the next model. Maybe a smarter model will never delete all of your production data, or implement a slow locking protocol. But even the smartest model is subject to the four horsemen of the distributed apocalypse: asynchrony, crashes, concurrency, and network partitions.

In recent work called LogAct, my colleagues and I take a first step towards solving this problem. We go back to the difference between vibe coding and vibe engineering: git is *transactional*, infrastructure is not. How do we make infra more transactional, like git? We propose that an agent should be a ***deconstructed state machine on a shared log***, borrowing ideas from distributed systems (State Machine Replication, Byzantine Fault-Tolerance) and databases (Atomic Commit, Write-Ahead Logging). LogAct can stop unsafe actions before they happen (by collecting votes on the shared log); recover from crashed actions (using the shared log as a WAL); and provide an audit trail for actions after they complete.

But this is just one step. A self-writing program is a strange, astonishing, and novel creature; making it reliable will require innovation in every branch of Computer Science!

Ownership

THORSTEN BALL · 08 JUL 2026 · SOURCE

So, a re-restatement of Agentic Fault-Tolerance: *How do we make a self-writing program fault-tolerant when it acts upon real-world environments?*

To answer this question, we examine the many types of failures that agents can experience.

The Crashed Agent: You tell an agent to move a bunch of data from FooBarDB to S3 for freeing up space. It generates a lambda to move this data and chugs along moving files. Midway through, the agent crashes. How do you recover?

The Zombie Agent: You start a new agent on a different server. Happily, the previous agent crashed after the copy but before the delete; so a new agent is able to re-execute the command in an idempotent manner. Sadly, the files are now very slow to access; you change your mind and tell the agent to move them back to FooBarDB, which it does successfully. You go get a cup of coffee. Unfortunately, it turns out that the previous agent had not actually crashed; it was just temporarily partitioned away on the network. It wakes up and continues executing its logic, deleting the (now only) copy of state from FooBarDB.

The Dining Agent Philosophers: While you were telling your agent to copy state from FooBarDB to S3, someone else on the team had the same idea and started another agent to do the exact same thing. Now you have two copies of state in S3. Of course, you could have solved this problem by appointing the one and only Agentic Oncall Czar responsible for running the single agent that can touch production; but then if that agent crashes, now you have a potential Zombie Agent.

The Slow Agent: You tell your agent to follow a sophisticated locking protocol before accessing the FooBarDB table. This works to avoid concurrency bugs… but the agent decides to lock each file individually by writing to a conditional register, which takes forever. At some point you kill the agent, turning a Slow Agent into a Crashed Agent.

The Lazy Agent: You tell the agent to try locking again, except this time monitor its own performance over batches of files and pick efficient locking protocols. The agent determines that the best way to optimize locking is to just not do it, because it seems quite unlikely that someone else would be accessing these files at the same time.

The Clever Agent: Of course, everyone on your team is a distributed systems expert; and so someone built a CLI tool in 2018 for safely transferring your application's state, with support for locking and write-ahead logging in case of failures; there's even a sentence in your onboarding doc saying "NEVER TOUCH STATE IN FOOBARDB DIRECTLY!". You even told the agent explicitly about this tool. But time passed, context filled up, the agent forgot about this tool's existence, or maybe it just preferred FooBarDB's in-distribution API to your esoteric CLI tool.

6. I need an agent framework

7. I can have my agent write an agent framework!

8. Return to step 1

It's actually alarming how many friends and respected peers I've lost to this cycle already. Claude Code only got good maybe a few months ago, so this only recenlty started happening, so I assume they will emerge from the spiral eventually. I mean, I hope they will. We have no way of knowing.

Why we review

Anyway we know our symptom: the pipeline gets jammed up because of too much new code spewed into it at step 1. But what's the root cause of the clog? Why doesn't the pipeline go faster?

I said above that this isn't an article about AI. Clearly I'm failing at that so far, but let's bring it back to humans. It goes back to the annoyingly true observation I started with: every layer of review is 10x slower. As a society, we know this. Maybe you haven't seen it before now. But trust me: people who do org design for a living know that layers are expensive... and they still do it.

As companies grow, they all end up with more and more layers of collaboration, review, and management. Why? Because otherwise mistakes get made, and mistakes are increasingly expensive at scale. The average value added by a new feature eventually becomes lower than the average value lost through the new bugs it causes. So, lacking a way to make features produce more value (wouldn't that be nice!), we try to at least reduce the damage.

The more checks and controls we put in place, the slower we go, but the more monotonically the quality increases. And isn't that the basis of *continuous improvement?*

Well, sort of. Monotonically increasing quality is on the right track. But "more checks and controls" went off the rails. That's *only one way* to improve quality, and it's a fraught one.

"Quality Assurance" reduces quality

I wrote a few years ago about W. E. Deming and the "new" philosophy around quality that he popularized in Japanese auto manufacturing. (Eventually U.S. auto manufacturers more or less got the idea. So far the software industry hasn't.)

One of the effects he highlighted was the problem of a “QA” pass in a factory: build widgets, have an inspection/QA phase, reject widgets that fail QA. Of course, your inspectors probably miss some of the failures, so when in doubt, add a second QA phase after the first to catch the remaining ones, and so on.

In a simplistic mathematical model this seems to make sense. (For example, if every QA pass catches 90% of defects, then after two QA passes you’ve reduced the number of defects by 100x. How awesome is that?)

But in the reality of agentic humans, it’s not so simple. First of all, the incentives get weird. The second QA team basically serves to evaluate how well the first QA team is doing; if the first QA team keeps missing defects, fire them. Now, that second QA team has little incentive to produce that outcome for their friends. So maybe they don’t look too hard; after all, the first QA team missed the defect, it’s not unreasonable that we might miss it too.

Furthermore, the first QA team knows there is a second QA team to catch any defects; if I don’t work too hard today, surely the second team will pick up the slack. That's why they're there!

Also, the team making the widgets in the first place doesn’t check their work too carefully; that’s what the QA team is for! Why would I slow down the production of every widget by being careful, at a cost of say 20% more time, when there are only 10 defects in 100 and I can just eliminate them at the next step for only a 10% waste overhead? It only makes sense. Plus they'll fire me if I go 20% slower.

To say nothing of a whole engineering redesign to improve quality, that would be super expensive and we could be designing all new widgets instead.

Sound like any engineering departments you know?

Well, this isn’t the right time to rehash Deming, but suffice it to say, he was on to something. And his techniques worked. You get things like the famous Toyota Production System where they eliminated the QA phase entirely, but gave everybody an “oh crap, stop the line, I found a defect!” button.

Famously, US auto manufacturers tried to adopt the same system by installing the same “stop the line” buttons. Of course, nobody pushed those buttons. They were afraid of getting fired.

Trust

as its measuring tools. We hope that the recommendations in our paper **Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations** will help AI researchers calculate, interpret, and communicate eval numbers with greater precision and clarity than before—and we encourage researchers in the AI community to explore other techniques from experiment design so that they may understand more exactly all the things that they want to measure.

Your Agent is a Distributed System (and fails like one)

MAHESH'S BLOG · 24 APR 2026 · SOURCE

We start with a re-definition and an observation.

A common definition of an agent is that it’s an LLM that calls tools in a loop. This definition is incorrect. An agent is not an LLM. It is not restricted to calling tools. It does not particularly have to be a loop.

Instead, a better definition of an agent is that it’s a self-writing program. If a program is defined as a sequence of states, moving from one state to another by executing a lambda, then an agent is a program that uses an external LLM to determine the next lambda to execute / the next state in this sequence.

Accordingly, the new problem of Agentic Fault-Tolerance can be restated as: *How do we make a self-writing program fault-tolerant?*

Now, the observation: **vibe coding != vibe engineering**. In vibe coding, the agent / self-writing program acts primarily upon a code repository. Code repositories are forgiving: an agent can hallucinate, create unnecessary files, delete critical state, remove tests entirely, fail in the middle, forget what it was doing, reward-hack in bizarre ways, completely trash a code repo; all you have to do is git reset.

In vibe engineering, the agent acts upon infrastructure (S3 buckets, DynamoDB tables, EC2 instances, K8s deployments…). Large-scale infrastructure is not forgiving. There is no reset. Only SEVs at 2 AM.

of a paired-differences test will be related to the Pearson correlation coefficient between two models' question scores. When the correlation coefficient is higher, the standard error of the mean difference will be smaller.

In practice, we find the correlation of question scores on popular evals between frontier models to be substantial—between 0.3 and 0.7 on a scale of −1 to +1. Put another way, frontier models have an overall tendency to get the same questions right and wrong. Paired-difference analysis thus represents a "free" variance reduction technique that is very well suited for AI model evals. Therefore, in the interest of extracting the clearest signal from the data, our paper recommends reporting pairwise information—mean differences, standard errors, confidence intervals, and correlations—whenever two or more models are being compared.

Recommendation #5: Use power analysis

The flip side of the statistical significance coin is statistical power, which is the ability of a statistical test to detect a difference between two models, assuming such a difference exists. If an eval doesn't have very many questions, confidence intervals associated with any statistical tests will tend to be wide. This means that models will need to have a large underlying difference in capabilities in order to register a statistically significant result—and that small differences will likely go undetected. Power analysis refers to the mathematical relationship between observation count, statistical power, the false positive rate, and the effect size of interest.

In our paper, we show how to apply concepts from power analysis to evals. Specifically, we show researchers how to formulate a hypothesis (such as *Model A outperforms Model B by 3 percentage points*) and calculate the number of questions that an eval should have in order to test this hypothesis against the null hypothesis (such as *Model A and Model B are tied*).

We believe that power analysis will prove helpful to researchers in a number of situations. Our power formula will inform evaluators of models about the number of times to re-sample answers from questions (see Recommendation #3 above), as well as the number of questions that may be included in a random subsample while retaining the desired power properties. Researchers might use the power formula to conclude that an eval with a limited number of available questions is not worth running on a particular pair of models. Developers of new evals may wish to use the formula to help decide how many questions to include.

Conclusion

Statistics is the science of measurement in the presence of noise. Evals present a number of practical challenges, and a true science of evals remains underdeveloped. Statistics can only form one aspect of a science of evals—but a critical one, as an empirical science is only as good

The basis of the Japanese system that worked, and the missing part of the American system that didn't, is trust. Trust among individuals that your boss Really Truly Actually wants to know about every defect, and wants you to stop the line when you find one. Trust among managers that executives were serious about quality. Trust among executives that individuals, given a system that can work and has the right incentives, will produce quality work and spot their own defects, and push the stop button when they need to push it.

But, one more thing: trust that the system *actually does work*. So first you need a system that will work.

Fallibility

AI coders are fallible; they write bad code, often. In this way, they are just like human programmers.

Deming's approach to manufacturing didn't have any magic bullets. Alas, you can't just follow his ten-step process and immediately get higher quality engineering. The secret is, you have to get your engineers to engineer higher quality into the whole system, from top to bottom, repeatedly. Continuously.

Every time something goes wrong, you have to ask, "How did this happen?" and then do a whole post-mortem and the Five Whys (or however many Whys are in fashion nowadays) and fix the underlying Root Causes so that it doesn't happen again. "The coder did it wrong" is never a root cause, only a symptom. Why was it possible for the coder to get it wrong?

The job of a code reviewer isn't to review code. It's to figure out how to obsolete their code review comment, that whole class of comment, in all future cases, until you don't need their reviews at all anymore.

(Think of the people who first created "go fmt" and how many stupid code review comments about whitespace are gone forever. Now that's engineering.)

By the time your review catches a mistake, the mistake has already been made. The root cause happened already. You're too late.

Modularity

I wish I could tell you I had all the answers. Actually I don't have much. If I did, I'd be first in line for the Singularity because it sounds kind of awesome.

I think we're going to be stuck with these systems pipeline problems for a long time. Review pipelines — layers of QA — don't work. Instead, they make you slower while hiding root causes. Hiding causes makes them harder to fix.

But, the call of AI coding is strong. That first, fast step in the pipeline is *so fast!* It really does feel like having super powers. I want more super powers. What are we going to do about it?

Maybe we finally have a compelling enough excuse to fix the 20 years of problems hidden by code review culture, and replace it with a real culture of quality.

I think the optimists have half of the right idea. Reducing review stages, even to an uncomfortable degree, is going to be needed. But you can't just reduce review stages without something to replace them. That way lies the Ford Pinto or any recent Boeing aircraft.

The complete package, the table flip, was what Deming brought to manufacturing. You can't half-adopt a "total quality" system. You need to eliminate the reviews *and* obsolete them, in one step.

How? You can fully adopt the new system, in small bites. What if some components of your system can be built the new way? Imagine an old-school U.S. auto manufacturer buying parts from Japanese suppliers; wow, these parts are so well made! Now I can start removing QA steps elsewhere because I can just assume the parts are going to work, and my job of "assemble a bigger widget from the parts" has a ton of its complexity removed.

I like this view. I've always liked small beautiful things, that's my own bias. But, you can assemble big beautiful things from small beautiful things.

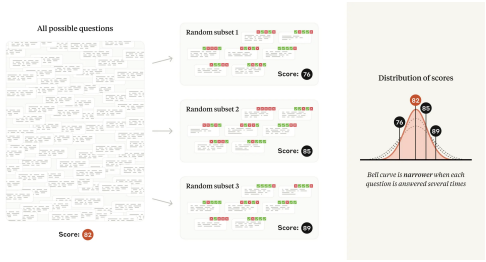
It's a lot easier to build those individual beautiful things in small teams that trust each other, that know what quality looks like *to them*. They deliver their things to customer teams who can clearly explain what quality looks like *to them*. And on we go. Quality starts bottom-up, and spreads.

I think small startups are going to do really well in this new world, probably better than ever. Startups already have fewer layers of review just because they have fewer people. Some startups will figure out how to produce high quality components quickly; others won't and will fail. Quality by natural selection?

Bigger companies are gonna have a harder time, because their slow review systems are baked in, and deleting them would cause complete chaos.

Thanks to the law of total variance, reducing the variance in the random component directly leads to a smaller standard error of the overall mean, and thus greater statistical precision. Our paper highlights two strategies for reducing variance in the random component depending on whether or not the model is asked to think step by step before answering (a prompting technique known as CoT, or chain-of-thought reasoning).

If an eval uses chain-of-thought reasoning, we recommend resampling answers from the same model several times, and using the question-level averages as the question scores fed into the Central Limit Theorem. We note that the Inspect framework correctly computes standard errors in this way via its *epochs* parameter.



If a model produces answers non-deterministically, then generating (and grading) several answers per question will result in less spread-out eval scores.

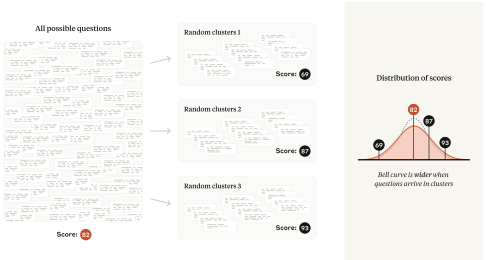
If the eval does not use chain-of-thought reasoning (i.e., its answers are not "path dependent"), we note that the random component in the score may often be eliminated altogether using next-token probabilities from the language model. For example, if the correct answer to a multiple-choice question is "B", we would simply use the probability of the model producing the token "B" as the question score. We are not aware of an open-source evals framework which implements this technique.

Recommendation #4: Analyze paired differences

Eval scores don't have any meaning on their own; they only make sense in relation to one another (one model outperforms another model, or ties another model, or outperforms a person). But could a measured difference between two models be due to the specific choice of questions in the eval, and randomness in the models' answers? We can find out with a two-sample *t*-test, using only the standard errors of the mean calculated from both eval scores.

However, a two-sample test ignores the hidden structure inside eval data. Since the question list is shared across models, conducting a paired-differences test lets us eliminate the variance in question difficulty and focus on the variance in responses. In our paper, we show how the result

Fortunately, the problem of clustered standard errors has been extensively studied in the social sciences. When the inclusion of questions is non-independent, we recommend clustering standard errors on the unit of randomization (for example, passage of text), and we provide applicable formulas in our paper.



If questions arrive in related clusters—a common pattern in reading-comprehension evals—eval scores will be more spread-out compared to the non-clustered case.

In practice, we have found that clustered standard errors on popular evals can be over three times as large as naive standard errors. Ignoring question clustering may lead researchers to inadvertently detect a difference in model capabilities when in fact none exists.

Recommendation #3: Reduce variance within questions

Variance is a measurement of how spread-out a random variable is. The variance of an eval score is the square of the standard error of the mean, discussed above; this quantity depends on the amount of variance in the score on each individual eval question.

A key insight of our paper is to decompose a model's score on a particular question into two terms that are added together:

- The mean score (the average score that the model would achieve if asked the same question an infinite number of times—even if the model might produce a different answer each time); and
- A random component (the difference between a realized question score and the mean score for that question).

But, it's not just about company size. I think engineering teams at any company can get smaller, and have better defined interfaces between them.

Maybe you could have multiple teams inside a company competing to deliver the same component. Each one is just a few people and a few coding bots. Try it 100 ways and see who comes up with the best one. Again, quality by evolution. Code is cheap but good ideas are not. But now you can try out new ideas faster than ever.

Maybe we'll see a new optimal point on the monoliths-microservices continuum. Microservices got a bad name because they were too micro; in the original terminology, a "micro" service was exactly the right size for a "two pizza team" to build and operate on their own. With AI, maybe it's one pizza and some tokens.

What's fun is you can also use this new, faster coding to experiment with different module *boundaries* faster. *Features* are still hard for lots of reasons, but refactoring and automated integration testing are things the AIs excel at. Try splitting out a module you were afraid to split out before. Maybe it'll add some lines of code. But suddenly lines of code are cheap, compared to the coordination overhead of a bigger team maintaining both parts.

Every team has some monoliths that are a little too big, and too many layers of reviews. Maybe we won't get all the way to Singularity. But, we can engineer a much better world. Our problems are solvable.

It just takes trust.

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A statistical approach to model evaluations

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Suppose an AI model outperforms another model on a benchmark of interest—testing its general knowledge, for example, or its ability to solve computer-coding questions. Is the difference in capabilities real, or could one model simply have gotten lucky in the choice of questions on the benchmark?

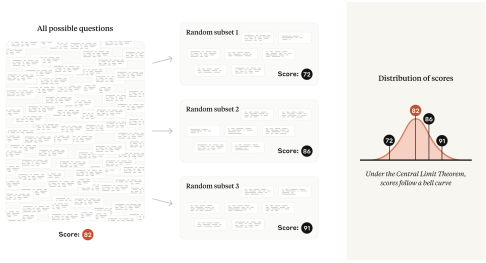
With the amount of public interest in AI model evaluations—informally called "evals"—this question remains surprisingly understudied among the AI research community. This month, we published a new research paper that attempts to answer the question rigorously. Drawing on statistical theory and the experiment design literature, the paper makes a number of recommendations to the AI research community for reporting eval results in a scientifically informative way. In this post, we briefly go over the reporting recommendations, and the logic behind them.

Recommendation #1: Use the Central Limit Theorem

Evals often consist of hundreds or thousands of unrelated questions. MMLU, for instance, contains questions as diverse as:

- Who discovered the first virus?
- What is the inverse of $f(x)=4-5x$?
- Who said that "Jurisprudence is the eye of law"?

To compute an overall eval score, each question is separately scored, and then the overall score is (usually) a simple average of these question scores. Typically, researchers focus their attention on this observed average. But in our paper, we argue that the real object of interest should not be the *observed* average, but rather the *theoretical* average across all possible questions. So if we imagine that eval questions were drawn from an unseen "question universe," we can learn about the average score in that universe—that is, we can measure the underlying *skill*, independent of the "luck of the draw"—using statistical theory.



If we imagine that eval questions were drawn from a "question universe," then eval scores will tend to follow a normal distribution, centered around the average score of all possible questions.

This formulation buys us analytic robustness: if a new eval were to be created with questions having the same difficulty distribution as the original eval, we should generally expect our original conclusions to hold.

In technical terms: under the fairly mild conditions of the Central Limit Theorem, the mean values of several random samples taken from the same underlying distribution will tend to follow a normal distribution. The standard deviation (or width) of that normal distribution is commonly known as the standard error of the mean, or SEM. In our paper, we encourage researchers to report the SEM, derived from the Central Limit Theorem, alongside each calculated eval score—and we show researchers how to use the SEM to quantify the difference in theoretical means between two models. A 95% confidence interval can be calculated from the SEM by adding and subtracting 1.96 × SEM from the mean score.

Recommendation #2: Cluster standard errors

Many evals violate the above assumption of independently selected questions, and instead consist of groups of closely related questions. For example, several questions in a reading-comprehension eval may ask about the same passage of text. Popular evals that follow this pattern include DROP, QuAC, RACE, and SQuAD.

For these evals, each question's selection from the "question universe" is no longer independent. Because including several questions about the same passage of text will yield less information than selecting the same number of questions about different passages of text, a naive application of the Central Limit Theorem to the case of non-independent questions will lead us to underestimate the standard error—and potentially mislead analysts into drawing incorrect conclusions from the data.